

Effects of watermelon supplementation on aortic blood pressure and wave reflection in individuals with prehypertension: a pilot study.

BACKGROUND: Oral L-citrulline is efficiently converted to L-arginine, the precursor for endothelial nitric oxide (NO) synthesis. Oral L-arginine supplementation reduces brachial blood pressure (BP). We evaluated the effects of watermelon supplementation on aortic BP and arterial function in individuals with prehypertension. **METHODS:** Heart rate (HR), brachial systolic BP (bSBP), brachial pulse pressure (bPP), aortic SBP (aSBP), aortic PP (aPP), augmentation index (AIx), AIx adjusted for HR of 75 beats/min (AIx@75), amplitude of the first (P1) and second (P2) systolic peaks, reflection time (Tr), and carotid-femoral pulse wave velocity (PWV) were evaluated in the supine position in nine subjects (four men/five women, age 54 ± 3 years) with prehypertension ($134/77 \pm 5/3$ mm Hg). Subjects were randomly assigned to 6 weeks of watermelon supplementation (L-citrulline/L arginine, 2.7 g/1.3 g/day) or placebo followed by a 4-week washout period and then crossover. **RESULTS:** There was a significant treatment effect (change in the value of watermelon minus placebo from baseline to 6 weeks) on bPP (-8 ± 3 mm Hg, $P < 0.05$), aSBP (-7 ± 2 mm Hg, $P < 0.05$), aPP (-6 ± 2 mm Hg, $P < 0.01$), AIx ($-6 \pm 3\%$, $P < 0.05$), AIx@75 ($-4 \pm 2\%$, $P < 0.05$), and P2 (-2 ± 1 mm Hg, $P < 0.05$). There was no significant treatment effect ($P > 0.05$) on bSBP, brachial diastolic BP (DBP), aortic DBP, Tr, P1, HR, and carotid-femoral PWV. **CONCLUSIONS:** This pilot study shows that watermelon supplementation improves aortic hemodynamics through a decrease in the amplitude of the reflected wave in individuals with prehypertension.